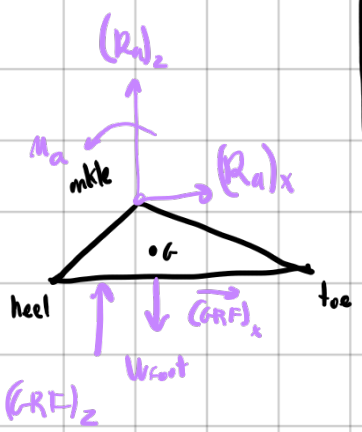
Human Body Project Part A

(1)



(2)

Foot

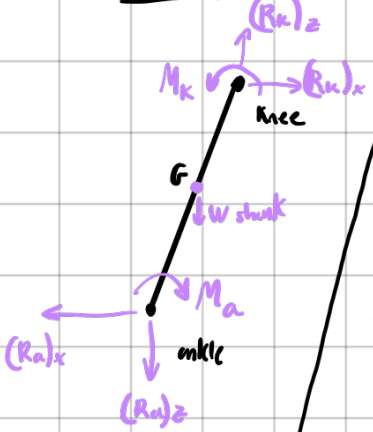


$$\sum F_x = m(a_G)_x \rightarrow (R_a)_x + (GRF)_x = m_F (a_{G_F})_x$$

$$\sum F_y = m(a_G)_z \rightarrow (R_a)_z - W_{foot} + (GRF)_z = m_F (a_{G_F})_z$$

$$\circlearrowleft \sum M_G = I_G \alpha \rightarrow M_a - (z_a - z_{G_F})(R_a)_x + (z_{G_F} - z_{COP})(GRF_x) - (x_G - x_a)(R_a)_z - (x_G - x_{cop})(GRF_z) = I_{G_F} \alpha_F$$

Shank

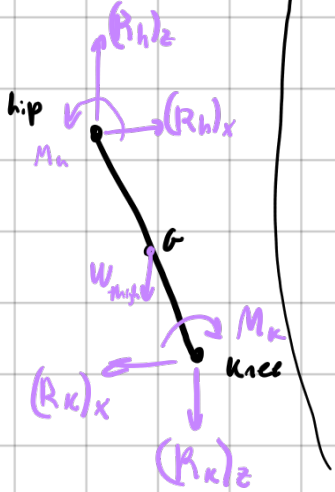


$$\sum F_x = m_{SH}(a_G)_x \rightarrow -(R_a)_x + (R_K)_x = m_{SH}(a_{G_{SH}})_x$$

$$\sum F_y = m_{SH}(a_G)_z \rightarrow -(R_a)_z + (R_K)_z - W_{SH} = m_{SH}(a_{G_{SH}})_z$$

$$\circlearrowleft \sum M_G = I_G \alpha \rightarrow -M_a + M_K + (x_K - x_{G_{SH}})(R_K)_z + (x_{G_{SH}} - x_a)(R_a)_z - (z_K - z_{G_{SH}})(R_K)_x - (z_{G_{SH}} - z_a)(R_a)_x = I_{G_{SH}} \alpha_{SH}$$

Thigh



$$\sum F_x = m_T(a_G)_x \rightarrow -(R_K)_x + (R_H)_x = m_T(a_{G_T})_x$$

$$\sum F_y = m_T(a_G)_z \rightarrow -(R_K)_z + (R_H)_z - W_T = m_T(a_{G_T})_z$$

$$\sum M_G = I_G \alpha \rightarrow -M_K + M_H - (x_{G_T} - x_H)(R_H)_z - (x_K - x_{G_T})(R_K)_z - (z_H - z_{G_T})(R_H)_x - (z_{G_T} - z_K)(R_K)_x = I_{G_T} \alpha_T$$

(3)

Foot

$(R_a)_x + (GRF)_x = m_f (a_{G_f})_x$

$(R_a)_x + (18.53) = 0.86(0.0328)$

$(R_a)_x = -18.5017\,N$

$(R_a)_z - W_{foot} + (GRF)_z = m_f (a_{G_f})_z$

$(R_a)_z - 0.86(9.81) + 595.9 = 0.86(0.3992)$

$R_{a_z} = -587.1201\,N$

$M_a - (z_a - z_{G_f})(R_a)_x + (z_{G_f} - z_{COP})(GRF_x) - (x_G - x_a)(R_a)_z - (x_G - x_{cop})(GRF_z) = I_{G_f}\alpha_f$

$M_a - (0.121 - 0.063)(-18.5017) + (0.063 - 0)(18.53) - (0.225 - 0.189)(-587.1201) - (0.225 - 0.307)(595.9) = 0.0037 \cdot (-2.8752)$

$M_a = -72.2513\,J$

Shank

$-(R_a)_x + (R_K)_x = m_{SH}(a_{G_{SH}})_x$

$(R_K)_x = 2.73(1.5369) - 18.5017$

$(R_K)_x = -14.3060\,N$

$-(R_a)_z + (R_K)_z - W_{SH} = m_{SH}(a_{G_{SH}})_z$

$(R_K)_z = 2.73(-0.2395) + 2.73(9.81) - 587.1201$

$(R_K)_z = -560.9926\,N$

$-M_a + M_K + (x_K - x_{G_{SH}})(R_K)_z + (x_{G_{SH}} - x_a)(R_a)_z - (z_K - z_{G_{SH}})(R_K)_x - (z_{G_{SH}} - z_a)(R_a)_x = I_{G_{SH}}\alpha_{SH}$

$M_K = -(72.2513 + (0.292 - 0.246)(-560.9926) + (0.246 - 0.189)(-587.1201) - (0.483 - 0.322)(-14.306) - (0.322 - 0.121)(-18.5017) - 0.0237(-5.6388))$

$M_K = -19.135\,J$

Thigh

$$-(R_K)_x + (R_H)_x = m_T (a_{GT})_x$$

$$14.306 + (R_H)_x = (8.92)(1.6365)$$

$$(R_H)_x = 0.2415\,N$$

$$-(R_K)_z + (R_H)_z - W_T = m_T (a_{GT})_z$$

$$(R_H)_z = (8.92)(-0.7012) - 560.9926 + 8.92(9.81)$$

$$(R_H)_z = -479.742\,N$$

$$-M_K + M_H - (x_{G_T} - x_H)(R_H)_z - (x_K - x_{GT})(R_K)_z - (z_H - z_{GT})(R_H)_x - (z_{GT} - z_K)(R_K)_x = I_{G_T}\alpha_T$$

$$M_H = -\Big[-M_K - (x_G - x_H)(-479.742) - (x_K - x_G)\underbrace{(-560.9926)}_{(R_H)_z} - (z_H - z_G)(0.2415) - (z_G - z_K)(-14.306) - (0.1903)(2.2077)\Big]$$

$$M_H = 12.4491\,J$$